

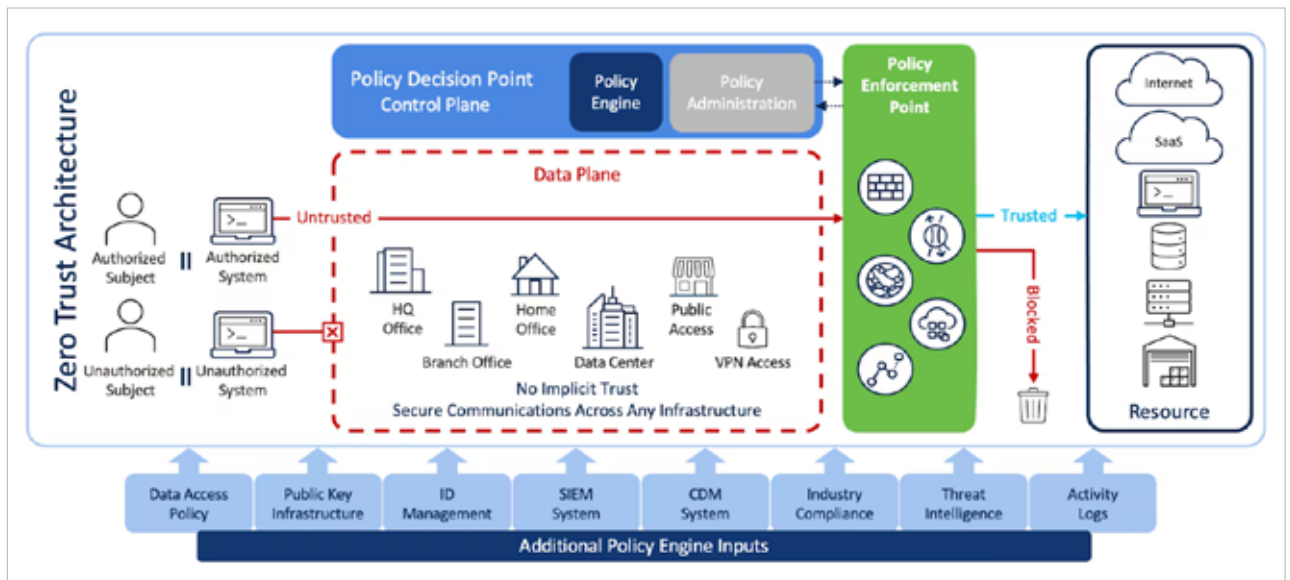


Figure 4: Zero trust architecture in Kubernetes security

```
securityContext:
  privileged: true
```

Automating threat response in a zero trust framework

An AI-driven IDS can trigger automated threat response by enabling integration with Kubernetes security policies such that threats are eliminated in real-time. Unlike current security controls based on static rule-based detection, AI-driven IDS learns over time and adapts based on new patterns of attack, thereby becoming more effective in detecting sophisticated threats. When based on a zero trust model, all network interactions are authenticated, reducing the likelihood of lateral movement by attackers.

Here's how AI empowers an automated threat response within a zero trust architecture.

Real-time anomaly detection:

Machine-learning-based IDS solutions (Snort, Suricata with ML) monitor workloads and identify unusual activity in real time.

Automated access control: Traffic is tagged by machine learning models and access controls are applied (e.g., Kubernetes network policies, admission controllers).

Real-time threat mitigation: Threats are detected by AI, which initiates immediate security measures such as blocking bad IPs, isolating infected pods, or enforcing new policies.

The automated response script is as follows:

```
import os
def block_ip(ip):
    os.system(f"kubectl annotate
namespace default netpol.blocked-
{ip}=deny'")
suspicious_ip = "192.168.1.100"
block_ip(suspicious_ip)
```

```
print(f"Blocked suspicious IP:
{suspicious_ip}")
```

By integrating eBPF, Snort, Suricata, and anomaly detection based on AI, businesses can develop an extremely effective real-time Kubernetes security monitoring system that provides strong cyber-attack protection. Intrusion detection by machine learning does not only expand detection rates but also minimises human-driven rule updates, thus enabling security automation to scale and learn more efficiently in contemporary cloud environments. 

References

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- <https://www.stamus-networks.com/hubfs/Library/Documents%20%28PDFs%29/StamusNetworks-WP-eBF-XDP-092021-1.pdf>
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